InsydeH2O® Eagle Stream 5.4 OEM Customization Guide

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Revision History

Revision Number	Description	Author	Release Date
0.1	Initial release.	James SX Lee	January, 2021
0.2	7 Adding Super IO • Fixed the path of SSDT table in section 7.1 and section 7.3 • Fixed the value of gSioGuid.PcdH2OSioAst2xxxChipModule in section 7.1 and section 7.3	James SX Lee	September, 2024

1 Introduction

We have created this document to assist engineers in developing InsydeH2O 5.4 BIOS for their Eagle Stream project. Please follow the Intel reference code for Eagle Stream chipset, CPU, and features porting. Eagle Stream InsydeH2O 5.4 BIOS has used it and we have tested it on the Eagle Stream CRB.

When using Eagle Stream releases from Insyde Software, we strongly suggest following these recommendations for OEM projects:

1. Update Intel reference code continuously and fix bugs.

2. Reference Eagle Stream CRB EagleStreamMultiBoardPkg code and then customize it for the OEM project.

The following terms are used throughout this document to describe varying aspects of input localization:

$(PROJECT_REL_PATH)	For CRB, it's Board/Intel For Project, it may be Board/Xyz
$(PROJECT_PKG)	For CRB, it's EagleStreamMultiBoardPkg For Project, it may be BoardNameBoardPkg
$(CHIPSET_REL_PATH)	Intel/EagleStream
$(CHIPSET_PKG)	EagleStreamChipsetPkg

2 Project Porting Rule

This section describes how to use OEM services and Override folders.

2.1 How to modify Kernel/Chipset code

If your project modifies kernel/chipset code, you need to feedback information regarding these modifications to the kernel/chipset team. Additionally, the modified code must be placed in the $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Override related location.

If a component is Eagle Stream OEM/ODM dependent, it is not necessary to put it into the Override folder.

InsydeH2O 5.4 treats Override folder as temp solution and on H2O 5.4 we use OEM services as customized method. Please feedback this override information to kernel/chipset team based on whether the code is for kernel or chipset. After reviewing, kernel/chipset will offer one new OEM services for your requirement, then you can put your modified code to this OEM services.

Clarification: OEM project engineers may face time constraints to produce code that functions well. To save time, engineers can simply override kernel/chipset code to indicate that there is no way for that code to serve their needs.

2.2 Modify Intel reference code and Feedback

Example 1: If you need to modify CrystalRidge.c of reference code. CrystalRidge.c, is defined in ServerSiliconPkg/Package.dsc as below: ServerSiliconPkg/Universal/Dxe/CrystalRidge/CrystalRidge.inf

Please follow these steps to override :

a) Copy **CrystalRidge.c** from $(CHIPSET_REL_PATH)/ServerSiliconPkg/Universal/Dxe/CrystalRidge to $(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/ServerSiliconPkg/Universal/Dxe/CrystalRidge

b) Add this code on $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc

```
!disable ServerSiliconPkg/Universal/Dxe/CrystalRidge/CrystalRidge.inf
ServerSiliconPkg/Universal/Dxe/CrystalRidge/CrystalRidge.inf {

  <SOURCE_OVERRIDE_PATH>

    $(PROJECT_PKG)/Override/Intel/EagleStream/ServerSiliconPkg/Universal/Dxe/CrystalRidge

}
```

Note: Please feedback this override to kernel/chipset, then they will review this code and decide whether or not to offer new OEM services.

3 Project Build

The CRB binary is supposed to build from $(PROJECT_REL_PATH)/BoardNameBoardPkg

Eagle Stream CRB BoardNameBoardPkg is EagleStreamMultiBoardPkg.

$(CHIPSET_REL_PATH) means Intel/EagleStream.

$(CHIPSET_PKG) means EagleStreamChipsetPkg.

$(PROJECT_REL_PATH) means "Board/Intel" on CRB, and "Board/Xyz" on project.

$(PROJECT_PKG) means EagleStreamMultiBoardPkg.

3.1 Separate schematics specific design into BoardNameBoardPkg

GPIO settings

IRQ routings

Flashmap, Ihisi protected area, etc

3.2 OEM should copy the CRB BoardNameBoardPkg as a reference

a) Chipset Code is put on EagleStreamChipsetPkg, ServerSecurityPkg, ServerSiliconPkg, ServerRasPkg and CpRcPkg. This code is maintained by the Chipset team.
OEM can use PCD/OEM services to do customization.
OEM services have 2 kinds: OemSvcKernelLib and OemSvcChipsetLib.

Supported OEM services Chipset Lib can be found on
$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Include/Library/BaseOemSvcChipsetLib.h
$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Include/Library/PeiOemSvcChipsetLib.h
$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Include/Library/DxeOemSvcChipsetLib.h
$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Include/Library/SmmOemSvcChipsetLib.h

b) Supported OEM services Kernel Lib can be found on
Insyde/InsydeOemServicesPkg/Include/Library/PeiOemSvcKernelLib.h
Insyde/InsydeOemServicesPkg/Include/Library/DxeOemSvcKernelLib.h
Insyde/InsydeOemServicesPkg/Include/Library/SmmOemSvcKernelLib.h

c) Build your project in the location below:
$(PROJECT_REL_PATH)/$(PROJECT_PKG)

Eagle Stream features a full clock that is used as H/W default. If it needs to change clock gen, you may use the following method:

- Use ME kit porting clock value.

4 Programming Platform Control Hub GPIO Pin

This section describes how to modify the GPIO settings according to the Platform control hub and OEM specifications

4.1 Introduction

To use Excel (or H2OIDE) to control the GPIO Settings by directly editing the GPIO Settings (.gpio) file. There is one GPIO Settings file for each of the boards. The files must have the name GPIO_XXXX.gpio, where XXXX matches one of the board names listed in [SkuIds] in Project.dsc.

4.2 Related files

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Include/ChipsetGpio.h

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/GpioCfgLib/GpioCfgLib.inf

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/ProgramGpioPei/ProgramGpioPei.inf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcChipsetLib/OemSvcGetGPIOTable.c

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcChipsetLib/OemSvcModifyGpioSettingTable.c

4.3 Setting

Modify the GPIO initial values at build time.

- Modify Gpio_XXXXXX.gpio

Update GPIO default setting dynamically for different SKUs.

- Modify OemSvcModifyGpioSettingTable.c

PCD GPIO setting value.

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Include/ChipsetGpio.h
typedef struct { UINT32 GpioPad; UINT8 PadMode;

```
  UINT8   HostSoftPadOwn;
  UINT8   Direction;
  UINT8   OutputState;
  UINT8   InterruptConfig;
  UINT8   PowerConfig;
  UINT8   ElectricalConfig;
  UINT8   LockConfig;
  UINT8   OtherSettings;
  UINT8   Rsvdbits;
} PCD_GPIO_CONFIG;
```

```
    $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc
[SkuIds]
  0x00|DEFAULT
  0x16|TypeArcherCityRP
  0x20|TypeArcherCityModular
  0x21|TypeArcherCityEVB
  0x22|TypeArcherCityXPV
  0x23|TypeBigPineKey
[LibraryClasses.common.PEIM]
  GpioCfgLib|$(PROJECT_PKG)/Library/GpioCfgLib/GpioCfgLib.inf
```

```
    $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/GpioCfgLib/GpioCfgLib.inf
[Defines]
  INF_VERSION              = 0x00010005
  BASE_NAME                = GpioCfgLib
  FILE_GUID                = c6b57539-2df4-40f7-a1c6-91a2bae359e7
```

```
  MODULE_TYPE                = PEIM
  VERSION_STRING             = 1.0
  LIBRARY_CLASS              = GpioCfgLib

[Sources]
# The file name is GPIO_xxxx.gpio, where xxxx matches [Skuids] in Project.dsc
  GPIO_DEFAULT.gpio
  GPIO_TypeArcherCityEVB.gpio
  GPIO_TypeArcherCityModular.gpio
  GPIO_TypeArcherCityRP.gpio
  GPIO_TypeArcherCityXPV.gpio
  GPIO_TypeBigPineKey.gpio
  GetGpioCfg.c
```

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/ProgramGpioPei/ProgramGpioPei.inf

```
EFI_STATUS
ProgramGPIOEntryPoint (
  …
  )
{
    …
    Status = GetGpioCfg(0, &PcdGpioTable, &GpioTableCount);
    …
    OemSvcGetGpioTable (&PcdGpioTable, &GpioTableCount);
    …
  Status = OemSvcModifyGpioSettingTable(GpioDefinitionTable);
  …
  Status = GpioConfigurePads (GpioTableCount, GpioDefinitionTable);
  …
```

```
  return EFI_SUCCESS;
}
```

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/ Library/PeiOemSvcChipsetLib/OemSvcGetGPIOTable.c

```
EFI_STATUS
OemSvcGetGpioTable(
  PCD_GPIO_CONFIG **GpioTable,
  UINT32          *PcdGpioTableCount
  )
{
  …
  GuidHob = GetFirstGuidHob (&gEfiPlatformInfoGuid);
  …
    PlatformInfo = GET_GUID_HOB_DATA (GuidHob);
    BoardId = PlatformInfo->BoardId;
   …
    Status = GetGpioCfg((UINT8) BoardId, GpioTable, PcdGpioTableCount);
  …
  return  EFI_MEDIA_CHANGED;
}
```

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcChipsetLib/OemSvcModifyGpioSettingTable.c

```
EFI_STATUS
OemSvcModifyGpioSettingTable (
  IN OUT  GPIO_INIT_CONFIG  *GpioTable
  )
```

```
{

//  UINT32                Data32;
//
//  Data32 = IoRead32 (PcdGet16(PcdPchGpioBaseAddress) + 0x60);
//  Data32 |= BIT30;
//  IoWrite32 (PcdGet16(PcdPchGpioBaseAddress) + 0x60, Data32);
//
  return EFI_MEDIA_CHANGED;
}
```

OEM services for GPIO

InsydeH2O 5.4 offers an OEM service "OemSvcModifyGpioSettingTable" for customers to change their GPIO setting before programming the PCD GPIO table setting to Chipset.

5 Adding CPU Microcode

Microcode updates consist of a processor-supplied binary that contains a descriptive header and data. The microcode update is composed of a multi-byte header, followed by the encoded data, and then followed by an optional extended signature table.

5.1 Introduction

OEM teams can follow the steps below to complete microcode integration:

1. Copy $(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode/Microcode.inf to $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode/Microcode.inf

2. Place the new Microcode.inc or TXT file on $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode

3. Modify $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode/Microcode.inf

```
[Sources]

  m_97_806f0_8a000370.inc  (add new one or upgrade on here)
```

4. Add this code on $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc

```
!disable $(CHIPSET_PKG)/Binary/Microcode/Microcode.inf

$(CHIPSET_PKG)/Binary/Microcode/Microcode.inf {

  <SOURCE_OVERRIDE_PATH>

    $(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode

}
```

5.2 Related files

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode/Microcode.inf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc

5.3 Setting

Microcode.inf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode/Microcode.inf

```
…
[Sources]
  m_97_806f0_8a000370.inc   #A0
```

Project.dsc

```
      $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc
```

```
…
   !disable $(CHIPSET_PKG)/Binary/Microcode/Microcode.inf
   $(CHIPSET_PKG)/Binary/Microcode/Microcode.inf {
     <SOURCE_OVERRIDE_PATH>
       $(PROJECT_PKG)/Override/$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/Binary/Microcode
   }
```

6 Programming Azalia with Verb Table

This section describes how to program Azalia with Verb Table.

6.1 Introduction

InsydeH2O 5.4 has an OEM service "OemSvcGetVerbTable" to provides the interface for customer to customize the Verb Table.

6.2 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcKernelLib/OemSvcGetVerbTable.c

6.3 Setting

OemSvcGetVerbTable.c

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcKernelLib/OemSvcGetVerbTable.c

…
//
// To define Verb Table ID.
//
#if 0
  // Sample code for VerbTable
  // HdaVerbTableAlc700
#define OEM_VERB_TABLE_ID_1 1
  //
  // VerbTable: (Realtek ALC700) CNL RVP
  // Revision ID = 0xff
  // Codec Verb Table for CNL PCH boards
  // Codec Address: CAd value (0/1/2)
  // Codec Vendor: 0x10EC0700

```
#define OEM_VERB_TABLE_1_HEADER1 0x10EC0700, \
                                 0x00000000, \
                                 0xFF, \
                                 0x01, \
                                 161, \
                                 0x0
#define OEM_VERB_TABLE_1_DATA1 0x001720F2, \
                               0x00172110, \
                               0x001722EC, \
                               0x00172310, \
...
```

Warning: The header includes many characteristics of an audio codec. Currently, only 2 fields of the header are examined to identify an audio codec:

1. Vendor ID / Device ID

2. Revision

If Revision is set to 0xFF, this means the revision will not be examined to identify an audio codec, but the Vendor ID / Device ID will still be examined.

7 Adding Super IO

This section describes how to add a Super IO device.

7.1 Introduction

Assume that the SIO name is XXXX in the following description.

1. Copy a required SIO package(SioAst2xxxPkg) into the root folder of your source code.
2. Import SIO package

   Add "!import SioAst2xxxPkg/Package.dsc" in Project.dsc.

   Add "!import SioAst2xxxPkg/Package.fdf" in Project.fdf.
3. If required, change the PCD settings of SioAst2xxxPkg:

   Add PCDs in Project.dsc and then modify them if required.

   A. Set corresponding value to change the name showed in setup.

      # PcdH2OSioAst2xxxChipModule | ChipModule

      # 0x00 | Aspeed 2300

      # 0x01 | Aspeed 2400

      # 0x02 | Aspeed 2500/2520

      # 0x03 | Aspeed 2600

      # 0x04-0xFF | Reserved

      gSioGuid.PcdH2OSioAst2xxxChipModule|0x03

   B. Turn auto detection off.

      gSioGuid.PcdH2OSioAst2xxxAutoDetectConfigPortSupported|FALSE

   C. Change the path of SSDT table. (it is LPC0 in Eagle Stream)

      gSioGuid.PcdH2OSioAst2xxxAslDeviceNamePrefix|"_SB.PC00.LPC0"

   D. Disable the ComDebugIoPei.inf by setting PCD.

      gSioGuid.PcdH2OSioAst2xxxChip1Uart1DebugPortSupported|FALSE

   E. Set the _PRS by PCD. (0x3F8 0x2F8 0x3E8 0x2E8 are open in default)

      gSioGuid.PcdH2OSioAst2xxxChip1UartResourceSetting|{

      UINT16(0x03F8), UINT8(0x08), UINT16(0x0CF8), UINT8(0x01), UINT8(0x08), UINT8(0x19), UINT8(0x08), UINT8(0x3F), \

```
UINT16(0x02F8), UINT8(0x08), UINT16(0x0CF8), UINT8(0x01), UINT8(0x08), UINT8(0x19),
UINT8(0x08), UINT8(0x3F), \

UINT16(0x02E8), UINT8(0x08), UINT16(0x0CF8), UINT8(0x01), UINT8(0x08), UINT8(0x19),
UINT8(0x08), UINT8(0x3F), \

UINT16(0x03E8), UINT8(0x08), UINT16(0x0CF8), UINT8(0x01), UINT8(0x08), UINT8(0x19),
UINT8(0x08), UINT8(0x3F) \

}
```

7.2 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.fdf

SIO package Sio/(SioXXXXPkg)/*.*

7.3 Setting

Project.dsc

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc
… !import SioXXXXPkg/Package.dsc … [PcdsFixedAtBuild] gSioGuid.PcdH2OSioAst2xxxChipModule|0x03 gSioGuid.PcdH2OSioAst2xxxAutoDetectConfigPortSupported|FALSE gSioGuid.PcdH2OSioAst2xxxAslDeviceNamePrefix|"_SB.PC00.LPC0" …

Project.fdf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.fdf
… !import SioAst2xxxPkg/Package.fdf …

8 PCISkipTable

This section explains where to set a skip table.

8.1 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/DxeOemSvcKernelLib/OemSvcInstallPciSkipTable.c

8.2 Setting

OemSvcInstallPciSkipTable.c

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/DxeOemSvcKernelLib/OemSvcInstallPciSkipTable.c

…
//
// module variables
//
PCI_SKIP_TABLE mPciSkipTable[] = {
  {
  0x8086,
  0xA224
  },
  {
  0xffff,
  0xffff
  }
};
…

9 Program SsidSvid

This section provides information on where to set a device's SsidSvid.

OEM can use a PcdDefaultSsidSvid to update SSID default value. Or do entire task to handle special case in OemSvcUpdatSsidSvidInfo.

9.1 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/DxeOemSvcKernelLib/OemSvcUpdateSsidSvidInfo.c

9.2 Setting

OemSvcUpdateSsidSvidInfo.c

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/DxeOemSvcKernelLib/OemSvcUpdateSsidSvidInfo.c
… /** The OemSvc is used to update SSID/SVID value by OEM. @param[in] Bus Bus number. @param[in] Dev Device number. @param[in] Func Function number. @param[in] VendorID Vendor ID. @param[in] DeviceID Device ID. @param[in] ClassCode Class Code. @param[in out] SsidSvid Pointer to SSID/SVID. @retval EFI_UNSUPPORTED Returns unsupported by default. @retval EFI_SUCCESS OEM handled SSID/SVID programming on this PCI device. Skip default kernel programming mechanism.

```
  @retval     EFI_MEDIA_CHANGED     Updated SsidSvid value and returned this value for default
                                    programming mechanism.

**/

EFI_STATUS

OemSvcUpdateSsidSvidInfo (

  IN     UINT8     Bus,

  IN     UINT8     Dev,

  IN     UINT8     Func,

  IN     UINT16    VendorId,

  IN     UINT16    DeviceId,

  IN     UINT16    ClassCode,

  IN OUT UINT32    *SsidSvid

  )

{

  /*++

    Todo:

      Add project specific code in here.

  --*/

  UINT8       Index;

  EFI_STATUS  Status;


  Status = EFI_UNSUPPORTED;

  for (Index = 0; SsidTable[Index].SpecialSsidSvidFunction != NULL; Index++ ) {

    if (SsidTable[Index].VendorId == VendorId) {

      if ((SsidTable[Index].DeviceId == DEVICE_ID_DONT_CARE)

        || (SsidTable[Index].DeviceId == DeviceId)){

        Status = SsidTable[Index].SpecialSsidSvidFunction (Bus, Dev, Func, SsidSvid);

        break;
```

```
      }
    }
  }

  return Status;
}
```

10 FlashMap

This section describes how to modify the BIOS size of description region.

10.1 Introduction

If BIOS image size does not match the BIOS size of description region, please do not use null binary image appending BIOS image to meet BIOS size of description region in your project, else it will cause BIOS R/W variable error during POST.

We suggest you to follow one of the two scenarios below:

1. Please adjust setting of description region to meet your BIOS image size.
2. Please modify Project.fdf [Defines] to adjust BIOS image size.

The default size of CRB System BIOS is 16 mega bytes (MB).

To modify Project.fdf [Defines] for project's requirement in the location below:

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/FlashRegionDefines.fdf

```
DEFINE FLASH_BASE = 0xFF000000
DEFINE FLASH_SIZE = 0x01000000
```

10.2 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/FlashRegionDefines.fdf

10.3 Setting

FlashRegionDefines.fdf (example)

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/FlashRegionDefines.fdf
… DEFINE FLASH_BASE = 0xFF000000 DEFINE FLASH_SIZE = 0x01000000 DEFINE BLOCK_SIZE = 0x00001000 DEFINE NUM_BLOCKS = $(FLASH_SIZE)/$(BLOCK_SIZE)

```
DEFINE FLASH_REGION_FVMAIN_SIZE  = 0x00999000 - $(FLASH_REGION_BIOSACM_SIZE) -
$(FLASH_REGION_FVAPEI_SIZE) - $(FLASH_REGION_FVUNSIGNED_SIZE) -
$(FLASH_REGION_BIOS_EVENT_STORAGE_SIZE) -
$(FLASH_REGION_INSYDE_BOOT_LOGO_SIZE) - $(FLASH_REGION_FV_CFR_SIZE)


DEFINE FLASH_REGION_FVUNSIGNED_SIZE                        = 0

!if gInsydeTokenSpaceGuid.PcdUnsignedFvSupported

DEFINE FLASH_REGION_FVUNSIGNED_SIZE                        = 0x00010000

!endif


DEFINE FLASH_REGION_FV_VARIABLE_DEFAULTS_SIZE               = 0x00040000

DEFINE FLASH_REGION_FLASH_DEVICE_MAP_SIZE                   = 0x00001000


!if gInsydeTokenSpaceGuid.PcdH2OBdsOemBadgingSupported

DEFINE FLASH_REGION_INSYDE_BOOT_LOGO_SIZE                    = 0x00000000

!else

DEFINE FLASH_REGION_INSYDE_BOOT_LOGO_SIZE                    = 0x00004000

!endif



DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_MICROCODE_SIZE      =
0x00180000

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_OEM_DMI_STORE_SIZE    =
0x00001000

#

# MSDM will be protected on BIOS Guard update.

# Insyde BIOS guard program unit is 64K,so address must be 64k aligent.

#  [Reference setting for the platform don't support BIOS Guard]

#
```

```
DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_MSDM_DATA_SIZE    =
0x00010000

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_BVDT_SIZE       =
0x00000200

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV1_SIZE       =
0x0000AE00

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV2_SIZE       =
0x00018000


!if $(CPU_SPR_SUPPORT) == TRUE

DEFINE FLASH_REGION_FV_CFR_SIZE                          = 0x00020000

!else

DEFINE FLASH_REGION_FV_CFR_SIZE                          = 0x00000000

!endif


DEFINE FLASH_REGION_BIOSACM_SIZE                          = 0

!if $(LTSX_SUPPORT) == YES

DEFINE FLASH_REGION_BIOSACM_SIZE                          = 0x00040000

!endif


DEFINE FLASH_REGION_FVAPEI_SIZE                          = 0

!if $(H2O_APEI_SUPPORT) == YES

DEFINE FLASH_REGION_FVAPEI_SIZE                          = 0x00030000

!endif


DEFINE FLASH_REGION_BIOS_EVENT_STORAGE_SIZE
      = 0x00000000

!if gH2OEventLogPkgTokenSpaceGuid.PcdH2OBiosEventStorageSupported

DEFINE FLASH_REGION_BIOS_EVENT_STORAGE_SIZE
      = 0x00006000
```

```
!endif


DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_VARIABLE_STORE_SIZE     =
0x00026000

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_WORKING_SIZE       =
0x00001000

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_SPARE_SIZE         = 0x00027000

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_FACTORY_COPY_SIZE      =
0x00004000

DEFINE FLASH_REGION_FV_BACKUP_SIZE                        = 0x00020000


!if $(FSP_BUILD) == TRUE

DEFINE FLASH_REGION_FV_FSP_S_SIZE                         = 0x00060000

  #

  # The space is not enough, so do not allocate DUMMYFV for now.

  # TODO: Wait for Intel to complete this feature before implementing it.

  #

DEFINE FLASH_REGION_FV_FSP_DUMMY_SIZE                      = 0 #0x000B8000

DEFINE FLASH_REGION_FV_FSP_M_SIZE                         = 0x00280000

DEFINE FLASH_REGION_FV_FSP_T_SIZE                         = 0x00008000

!else

DEFINE FLASH_REGION_FV_FSP_S_SIZE                         = 0

DEFINE FLASH_REGION_FV_FSP_DUMMY_SIZE                      = 0

DEFINE FLASH_REGION_FV_FSP_M_SIZE                          = 0

DEFINE FLASH_REGION_FV_FSP_T_SIZE                         = 0

!endif


DEFINE FLASH_REGION_FV_FSP_SIZE                           =
$(FLASH_REGION_FV_FSP_S_SIZE) + $(FLASH_REGION_FV_FSP_DUMMY_SIZE) +
$(FLASH_REGION_FV_FSP_M_SIZE) + $(FLASH_REGION_FV_FSP_T_SIZE)
```

```
DEFINE FLASH_REGION_FV_RECOVERY_SIZE                        = 0x00400000 -
$(FLASH_REGION_FV_FSP_SIZE)


#

# --------------------------   Region Offset Defined  ------------------------------

#

DEFINE FLASH_REGION_FVMAIN_OFFSET                           = 0x00000000

DEFINE FLASH_REGION_FVUNSIGNED_OFFSET                           =
$(FLASH_REGION_FVMAIN_OFFSET) + $(FLASH_REGION_FVMAIN_SIZE)

DEFINE FLASH_REGION_FV_VARIABLE_DEFAULTS_OFFSET                 =
$(FLASH_REGION_FVUNSIGNED_OFFSET) + $(FLASH_REGION_FVUNSIGNED_SIZE)

DEFINE FLASH_REGION_FLASH_DEVICE_MAP_OFFSET                     =
$(FLASH_REGION_FV_VARIABLE_DEFAULTS_OFFSET) +
$(FLASH_REGION_FV_VARIABLE_DEFAULTS_SIZE)

DEFINE FLASH_REGION_INSYDE_BOOT_LOGO_OFFSET                     =
$(FLASH_REGION_FLASH_DEVICE_MAP_OFFSET) +
$(FLASH_REGION_FLASH_DEVICE_MAP_SIZE)

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_MICROCODE_OFFSET     =
$(FLASH_REGION_INSYDE_BOOT_LOGO_OFFSET) +
$(FLASH_REGION_INSYDE_BOOT_LOGO_SIZE)

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_OEM_DMI_STORE_OFFSET  =
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_MICROCODE_OFFSET) +
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_MICROCODE_SIZE)

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_MSDM_DATA_OFFSET   =
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_OEM_DMI_STORE_OFFSET) +
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_OEM_DMI_STORE_SIZE)

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_BVDT_OFFSET      =
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_MSDM_DATA_OFFSET) +
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_MSDM_DATA_SIZE)

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV1_OFFSET      =
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_BVDT_OFFSET) +
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_BVDT_SIZE)

DEFINE FLASH_REGION_FV_CFR_OFFSET                           =
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV1_OFFSET) +
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV1_SIZE)
```

```
DEFINE FLASH_REGION_BIOSACM_OFFSET                          =
$(FLASH_REGION_FV_CFR_OFFSET) + $(FLASH_REGION_FV_CFR_SIZE)

DEFINE FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV2_OFFSET      =
$(FLASH_REGION_BIOSACM_OFFSET) + $(FLASH_REGION_BIOSACM_SIZE)

DEFINE FLASH_REGION_FVAPEI_OFFSET                           =
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV2_OFFSET) +
$(FLASH_REGION_NV_COMMON_STORE_SUBREGION_NV_RSV2_SIZE)

DEFINE FLASH_REGION_BIOS_EVENT_STORAGE_OFFSET
     = $(FLASH_REGION_FVAPEI_OFFSET) + $(FLASH_REGION_FVAPEI_SIZE)

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_VARIABLE_STORE_OFFSET   =
$(FLASH_REGION_BIOS_EVENT_STORAGE_OFFSET) +
$(FLASH_REGION_BIOS_EVENT_STORAGE_SIZE)

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_WORKING_OFFSET     =
$(FLASH_REGION_NVSTORAGE_SUBREGION_NV_VARIABLE_STORE_OFFSET) +
$(FLASH_REGION_NVSTORAGE_SUBREGION_NV_VARIABLE_STORE_SIZE)

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_SPARE_OFFSET         =
$(FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_WORKING_OFFSET) +
$(FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_WORKING_SIZE)

DEFINE FLASH_REGION_NVSTORAGE_SUBREGION_NV_FACTORY_COPY_OFFSET     =
$(FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_SPARE_OFFSET) +
$(FLASH_REGION_NVSTORAGE_SUBREGION_NV_FTW_SPARE_SIZE)


DEFINE FLASH_REGION_FV_BACKUP_OFFSET                          =
$(FLASH_REGION_FV_FSP_S_OFFSET) - $(FLASH_REGION_FV_BACKUP_SIZE)

DEFINE FLASH_REGION_FV_FSP_S_OFFSET                           =
$(FLASH_REGION_FV_FSP_M_OFFSET) - $(FLASH_REGION_FV_FSP_S_SIZE)

DEFINE FLASH_REGION_FV_FSP_M_OFFSET                           =
$(FLASH_REGION_FV_FSP_T_OFFSET) - $(FLASH_REGION_FV_FSP_M_SIZE)

DEFINE FLASH_REGION_FV_FSP_T_OFFSET                           =
$(FLASH_REGION_FV_RECOVERY_OFFSET) - $(FLASH_REGION_FV_FSP_T_SIZE)

DEFINE FLASH_REGION_FV_RECOVERY_OFFSET                        = $(FLASH_SIZE) -
$(FLASH_REGION_FV_RECOVERY_SIZE)


DEFINE FLASH_REGION_FV_FSP_OFFSET                             =
$(FLASH_REGION_FV_RECOVERY_OFFSET) - $(FLASH_REGION_FV_FSP_SIZE)

...
```

11 HotKey

This section describes how to modify the hotkey to fit in the OEM project feature.

11.1 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.dsc

Insyde/InsydeModulePkg/InsydeModulePkg.dec

Insyde/InsydeModulePkg/Package.dsc

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/FrontPageDxe/FrontPageDxe.inf

Insyde/InsydeModulePkg/Universal/BdsDxe/BdsDxe.inf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/DxeOemSvcKernelLib/OemSvcInstallPostKeyTable.c

11.2 Setting

Insyde/InsydeModulePkg/InsydeModulePkg.dec

Insyde/InsydeModulePkg/InsydeModulePkg.dec

```
…
[PcdsDynamicEx]
  …
  #
  # BDS Hot Key default PCD. These PCDs will format as
  #   typedef struct _PCD_H2O_HOT_KEY {
  #     EFI_GUID HotKeyProtocolGuid;
  #     UINT32   Id;
  #   //CHAR8    HotKeys[];
  #   //CHAR8    Options[];
  #   //CHAR8    DevicePath[];
  #   } PCD_H2O_HOT_KEY;
  #
  gH2OBdsHotKeyGuid.PcdH2OHotKeyDummy|{0}|VOID*|0x00000001
  gH2OBdsHotKeyGuid.PcdH2OHotKeyFrontPage|{0}|VOID*|0x00000002
```

Insyde/InsydeModulePkg/Package.dsc

```
Insyde/InsydeModulePkg/Package.dsc
...
[PcdsDynamicExDefault]
  ...
  !if gInsydeTokenSpaceGuid.PcdFrontPageSupported
  !if gInsydeTokenSpaceGuid.PcdDynamicHotKeySupported == FALSE
    gH2OBdsHotKeyGuid.PcdH2OHotKeyFrontPage|{GUID("00000000-0000-0000-0000-
  000000000000"), UINT32(0), "esc", "", "VenHw(E0BA9B98-DD2D-4434-BB94-
  599CC9E4305D)"}

  gH2OBdsHotKeyDescGuid.PcdH2OHotKeyFrontPageDesc|{STRING_TOKEN(STR_OEM_BADGING
  _STR_ESC), STRING_TOKEN(STR_OEM_BADGING_STR_ESC_SELECT),"x=0em;y=0em;
  xorigin=left;yorigin=bottom;"}

  gH2OBdsHotKeyDescQuietGuid.PcdH2OHotKeyFrontPageDescQuiet|{STRING_TOKEN(STR_OEM
  _BADGING_STR_ESC), STRING_TOKEN(STR_OEM_BADGING_STR_ESC_SELECT),"x=0em;
  y=0em;xorigin=center;yorigin=top(80%);"}
  !endif
 !endif
```

$(PROJECT_REL_PATH)/$(CHIPSET_PKG)/FrontPageDxe/FrontPageDxe.inf

```
$(PROJECT_REL_PATH)/$(CHIPSET_PKG)/FrontPageDxe/FrontPageDxe.inf
...
H2O_BDS_LOAD_OPTION_VENDOR_DEVICE_PATH      gH2OFrontPageDevicePath = {
 {
  {
    HARDWARE_DEVICE_PATH,
    HW_VENDOR_DP,
    {
      (UINT8) (sizeof (VENDOR_DEVICE_PATH)),
      (UINT8) ((sizeof (VENDOR_DEVICE_PATH)) >> 8)
```

```
      }
    },
    H2O_BDS_LOAD_OPTION_FRONT_PAGE_GUID
  },
  {
    END_DEVICE_PATH_TYPE,
    END_ENTIRE_DEVICE_PATH_SUBTYPE,
    {
      (UINT8) (END_DEVICE_PATH_LENGTH),
      (UINT8) ((END_DEVICE_PATH_LENGTH) >> 8)
    }
  }
};
```

Insyde/InsydeModulePkg/Universal/BdsDxe/BdsDxe.inf

Insyde/InsydeModulePkg/Universal/BdsDxe/BdsDxe.inf

```
…
PcdHotKeyEnumeration (
…
) {
  …
  Status = GetTokenNumList (&gH2OBdsHotKeyGuid, &TokenList, &TokenCount);
  …
  for (Index = 0; Index < TokenCount; Index++) {
      HotKeyOption = PcdHotKeyCreateOption (TokenList[Index]);
      …
  }
  …
  return;
```

```
}
```

12 Modify RC Policy

This section describes how to set the RC policy to fit in the OEM project feature.

12.1 Introduction

The policy drivers configure chipset settings in the Intel Reference Code.

1). Insyde Chipset Policy is in the location below:

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/PolicyInit/InsydeChipsetPolicy

It involves drivers as below.

GopPolicyDxe/GopPolicyDxe.inf

2). Each policy default value is setting by setup variable or PCD. OEM can modify it by your project requirement.

3). Each PPI or Protocol have a PEI/DXE OemSvcChipsetLib function for OEM to customize. This function is called before installing the PPI or Protocol.

12.2 Related files

$(CHIPSET_REL_PATH)/$(CHIPSET_PKG)/PolicyInit/InsydeChipsetPolicy/*.*

12.3 Setting

OEM library for RC policy

There are OEM library for RC code parameters changed in

- $(CHIPSET_REL_PATH)/ServerPlatformPkg/Library/OemCrystalRidgeLib/OemCrystalRidgeHooks.inf
- $(CHIPSET_REL_PATH)/ServerPlatformPkg/Library/OemIioInit/OemIioInitHooks.inf
- $(CHIPSET_REL_PATH)/ServerPlatformPkg/Library/OemProcessorStartupLib/OemProcessorStartupLib.inf
- $(CHIPSET_REL_PATH)/ServerPlatformPkg/Library/OemProcMemInitLib/OemProcMemInitLib.inf
- $(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/OemRcLib/OemRasLib/OemRasLib.inf

13 Skip Boot Mode

This section describes how to skip boot mode detection.

13.1 Introduction

How do I set boot mode to always be in “cold boot” mode when I bring up board on my project?

1). Set boot mode as BOOT_WITH_FULL_CONFIGURATION.

In OemSvcChangeBootMode () routine of OemSvcChangeBootMode.c file

*SkipPriorityPolicy == TRUE

*BootMode = BOOT_WITH_FULL_CONFIGURATION

Return EFI_SUCCESS

2). OemSvcChangeBootMode.c is in below location

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcKernelLib/OemSvcChangeBootMode.c

13.2 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcKernelLib/OemSvcChangeBootMode.c

13.3 Setting

OemSvcChangeBootmode.c

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Library/PeiOemSvcKernelLib/OemSvcChangeBootMode.c
… /** This OemService is queried to assign the default boot mode. OEM can use this service to assign default boot mode, and use the parameter SkipPriorityPolicy to control normal boot mode identification. If SkipPriorityPolicy be set as TRUE, the final boot mode is the same as the parameter "bootmode" which is assigned by this service. @param[in, out] *BootMode Assign default boot mode.

```
  @param[in, out]  *SkipPriorityPolicy If SkipPriorityPolicy == TRUE, then normal boot mode
identification will be skipped.


  @retval         EFI_UNSUPPORTED     Returns unsupported by default.

  @retval         EFI_SUCCESS         The service is customized in the project.

  @retval         EFI_MEDIA_CHANGED   The value of IN OUT parameter is changed.

  @retval         Others              Depends on customization.

**/

EFI_STATUS

OemSvcChangeBootMode (

  IN OUT EFI_BOOT_MODE                *BootMode,

  OUT    BOOLEAN                      *SkipPriorityPolicy

  )

{

   …

   PchPwrmBase = PmcGetPwrmBase ();

   PmConB = MmioRead32 (PchPwrmBase + R_PMC_PWRM_GEN_PMCON_B);


   if (PmConB & B_PMC_PWRM_GEN_PMCON_B_RTC_PWR_STS) {

     *BootMode = BOOT_WITH_DEFAULT_SETTINGS;

     // Also destroy the CmosCheckSum to avoid incorrect boot mode after reboot from RC code.

     Data16 = ReadCmos16 (CmosCheckSum2E);

     Data16++;

     WriteCmos16 (CmosCheckSum2E, Data16);

     return EFI_MEDIA_CHANGED;

   }

    …

    //
```

```
    // return EFI_SUCCESS only when setting *SkipPriorityPolicy = TRUE
    //

    return EFI_UNSUPPORTED;
}
```

14 Logo

This section describes how to add an OEM logo during POST.

14.1 Related files

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/FlashRegionDefines.fdf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.fdf

14.2 Setting

FlashRegionDefines.fdf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/FlashRegionDefines.fdf
… DEFINE FLASH_REGION_INSYDE_BOOT_LOGO_SIZE = 0x00004000 … DEFINE FLASH_REGION_INSYDE_BOOT_LOGO_OFFSET = $(FLASH_REGION_FLASH_DEVICE_MAP_OFFSET) + $(FLASH_REGION_FLASH_DEVICE_MAP_SIZE) …

Project.fdf

$(PROJECT_REL_PATH)/$(PROJECT_PKG)/Project.fdf
… # # Region: gH2OFlashMapRegionLogoGuid # Logo initializer: Logo(posX, posY, originX, originY[, id[, format]]) # # posX, posY, originX, originY, id and format need to use decimal. # # posX !VInteger that specifies the horizontal position of the logo, relative to the

```
#       origin specified by originX.
#
# posY !VInteger that specifies the vertical position of the logo, relative to the
#       origin specified by origin.
#
# originX - Enumerated value that specifies the horizontal origin of the logo.
#          0 = left, 1 = right, 2 = center. Default value is 2.
#
# originY - Enumerated value that specifies the vertical origin of the logo.
#          0 = top, 1 = bottom, 2 = center. Default value is 2.
#
# format !VOptional enumerated value that specifies the format of the logo.
#         0 = BMP, 1 = JPEG, 2 = GIF, 3 = PNG. If format is not specified,
#         then the tool will attempt to determine the fiel type automatically.
#
# id !VOptional enumerated value that specifies the logo type.
#      0 = boot logo, 1 = CPU badge logo, 2 = chipset badge logo, 255 = OEM. Default value is 0.
#
# posX, posY: Support pixels and percentage.
# Example:
#   Pixel: 100
#   Percentage: 20% (Percentange of resolution)
#
$(FLASH_REGION_INSYDE_BOOT_LOGO_OFFSET)|$(FLASH_REGION_INSYDE_BOOT_LOGO_SIZE
)
Alignment = 4096
Resize = Fixed
RegionType = gH2OFlashMapRegionLogoGuid | Logo (0, 19%, 2, 0)
```

```
Boards = 0

FILE = InsydeModulePkg/Universal/Console/Logo/InsydeBoot/InsydeBoot.jpg

…
```

Appendix A - Dual and Quad Output Read

This appendix describes how to enable Dual and Quad Output Read.

A.1 Prerequisites

To enable Dual and Quad Output Read, INTEL Flash Image Tool (FITm) is required.

A.2 Enabling Dual Output Read

There is a way to enable Dual Output Read.

If your Serial Flash Parts support 3Bh command, please follow the steps provided below:

- Execute FITm.
- Set ”Flash Configuration”->”Fast read support” to “Supported”.
- Set ”Flash Configuration”->”Dual Output Fast Read Support” to “Supported”.

Appendix B - H2ODDT USB3.0 Setup

This appendix describes how to enable H2ODDT by USB3.0 cable.

B.1 Prerequisites

New kernel, debug application, silicon, and USB3.0 debug cable are required to support USB3.0 debug:

1. New version of H2O Kernel: Skylake.05.04.34.0003 is verified.
2. New version of H2ODDT application: 3.4.9.3 is verified.
3. CRB and Chipset:
4. SKL-Y LPDDR3 RVP3 CRB is verified to support USB3.0 debug.
5. USB ports: All USB3.0 ports in back panel are available both on SKL-Y LPDDR3 RVP3 CRB.
6. USB3.0 Debug Cable: DataPro 1696-03C USB 3.0 'A' M/M DEBUG CABLE
7. Insyde USB2.0 Debug Cable

B.2 Setup Procedure

1. Modify H2O_DDT_DEBUG_IO from "Usb" to "Xhc" in project.env to build BIOS for H2ODDT USB3.0 debug interface.
2. Download newer H2ODDT application which supports USB3.0.
3. H2ODDT can be downloaded from X:/Tool/Common/H2ODDT (Developer Debug Tool)
4. Power up CRB and you can see POST code D0h/0Dh.
5. Connect USB3.0 debug cable on target platform and host machine.
6. Connect Insyde USB 2.0 debug cable on host machine. Insyde H2ODDT driver will check Insyde Debug Cable before using USB 3.0 debug interface.
7. After you finish step 4&5, you can see a new USB 3.0 cable in Window Device Manager.
8. Install USB 3.0 driver from H2ODDT application.
9. You can see POST code changes to D1h and start debugging.

B.3 Trouble Shooting

If the USB3.0 host controller of your host machine is Renesas chip, please disable the USB3.0 power management on it and try again. The utility(nusb3ult.exe) for disabling power management can be downloaded from:

X:/PCG/CRB/Skylake/Tool/Renesas USB 3.0 Host Controller Utility/ USB_Renesas_2.0.26.0_W7x64.7z

Appendix C - H2ODDT COM PORT Setup

This appendix describes how to enable H2ODDT by com port cable.

C.1 Prerequisites

Null modem cable are required to support com port H2ODDT debug:

1. New version of H2O Kernel 5.2 is verified.
2. New version of H2ODDT application: 3.4.9.3 is verified.
3. CRB and Chipset: WHL U CRB are verified to support com port debug.
4. Intel CRB support com port from BMC UART1.

C.2 Com port from PCH UART2 Setup Procedure

1. Debug interface.
2. Modify the "Insyde\InsydeModulePkg\H2ODebug\DebugConfig.exe".

   Change com port type = I/O, Address = 3F8. This address is BMC UART1 address. Build BIOS with "nmake uefi64ddt"
3. Connect Null modern cable to target platform and host machine.
4. Power on.
5. Start H2ODDT application and change project setting from USB to Serial.